MTB Business Lending
LLM Underwriting & Quick App
Complete Implementation Guide

Version 1.0 | October 2026 | Design / proof-of-concept specification | NOT an approved credit policy

Source basis: user-provided screenshots of the account and lending entity model, field mapping, underwriting strategy, decline reasons, product definitions, and earlier Query 1 / Query 2 / Query 4 design guide. The original workbook and approved policy documents were not provided; ambiguous rules are explicitly flagged for confirmation.

1. Executive architecture and decision boundaries

Pipeline: Excel Query 1 / Query 2 / Query 4 -> schema and relationship validation -> canonical borrower/loan model -> feature engineering -> versioned deterministic policy engine -> explanation LLM -> decision evidence and authorized review.

• The policy engine determines the simulated routing outcome; the LLM must not change it.

• Use three separate outputs: eligibility (Quick App / full underwriting / ineligible), credit assessment (candidate / decline candidate / review), and authorization (unapproved until an authorized process signs off).

• In a POC, label results SIMULATED and do not issue a binding approval, denial, adverse-action notice, or credit offer.

• A missing key, missing credit report, stale data, unverified policy, or ambiguous relationship results in DATA_REVIEW / POLICY_REVIEW, not an invented decision.

2. Canonical entity relationship model

Entity	Keys / relationships	Purpose
Account	account_id PK	Business or individual customer; one Account to many Loans and CreditReports.
LLC_BI_LOAN__C	loan_id PK; account_id FK	Loan and requested product; loan to many LCResponse; optional one-to-one LoanExtension.
LLC_BI_LEGAL_ENTITIES__C	entity_involvement_id PK; loan_id FK; account_id FK	Junction supporting many-to-many Account-to-Loan and borrower / guarantor roles.
NFUSE_CREDIT_REPORT__C	credit_report_id PK; account_id FK	One Account to many reports; select correct borrower/guarantor and report date.
LOAN_EXTENSION__C	loan_extension_id PK; loan_id FK UNIQUE	Behavioral underwriting program and extension attributes.
SCOC_LCRESPONSE__C	lcresponse_id PK; loan_id FK	SBSS response records; one loan to many responses.
SCOC_LCRENTITY__C	lcrentity_id PK; lcresponse_id FK; account_id FK; entity_involvement_id FK	Connects scored entities to response and loan involvement.
SCOC_LCRDATASOURCEANALYSIS__C	analysis_id PK; lcrentity_id FK	One LCREntity to many source analyses.
SCOC_LCCHARACTERISTIC__C	characteristic_id PK; analysis_id FK	Key/value SBSS credit characteristics.
SCOC_LCINDICATOR__C	indicator_id PK; analysis_id FK	Key/value SBSS indicators, e.g. thinFileIndicator=Y.

Entity	Keys / relationships	Purpose
MTB_REGULATION__C	regulation_id PK; loan_id FK	Communicated credit decision record; historical evidence, not a model prediction.

Important relationship nuance: The screenshot describes a direct Account-to-Loan 1:N relationship AND a many-to-many association via the legal-entities junction. Keep both: primary account on loan, and all participating borrower/guarantor accounts in the junction. Do not assume every joined row is the primary borrower.

3. Excel file mapping and join contract

Input	Required columns / linkage	Transformation
Query 1 - master	loan_id, primary account_id, requested amount, product, product type/line, account demographics, financials, group credit, SBSS response / score when available	One canonical row per loan_id. Deduplicate account/loan and attach most relevant score by loan and response date.
Query 2 - indicator	analysis_id, lcrentity_id, lcresponse_id, loan_id/account_id through relationships, indicator_key, indicator_value, borrower_type	Pivot key/value per analysis or scored entity; retain raw rows, source IDs, and conflicts.
Query 4 - bureau	credit_report_id, account_id, report date, credit score, freeze, public records, bankruptcies, charge-offs, liens, OFAC, past due, collection	Join by account_id and select appropriate recent report per entity; never join on row order.

Query 2 screenshots show fields including SCOC__IndicatorKey__c, SCOC__IndicatorValue__c, SCOC__SBSS7Score__c, LLC_BI__Borrower_Type__c, and relationships through DataSourceAnalysis, LCREntity and LCResponse. Query 4 uses nFUSE__Credit_Score__c, nFUSE__Credit_Report_Freeze__c, nFUSE__OFAC_Alert_Indicator__c, and related credit report fields. API names must be checked against the source workbook before deployment.

4. Canonical data quality and identity validation

• Check uniqueness of primary keys; verify foreign-key referential integrity across all three exports.

• Require a primary loan account; use legal-entity involvement to identify Borrower vs Guarantor and assess each relevant report independently.

• Require effective dates for report, score and underwriting event; define approved recency limits.

• Normalize numeric money, credit scores, dates, Boolean Y/N and nulls; retain raw source value and normalized value.

• Reject conflicting values for the same key at the same source timestamp; do not silently take first row.

• Document source-to-target mapping, join cardinality, null rate, duplicate count and validation outcome per loan.

5. Product eligibility and routing (source-derived)

Rule / product	Source-derived condition	Proposed routing
Quick App	New money <= $100M and Tier I Group Credit <= $250M; NFPs excluded	Quick App eligible only if all eligibility and data checks pass.
Express BALOC	Unsecured; <= $100M. Increase over $100M may still use Quick App if total new money < $100M and group credit < $250M (as written in screenshot).	Route to product-specific check; clarify existing vs incremental exposure.

Rule / product	Source-derived condition	Proposed routing
Conventional BALOC	Over $100M must be secured.	Collateral review / full underwriting as required.
BIL	Unsecured term debt <= $100M; acceptance letter required.	Document check; potential conversion to term loan when collateral taken.
SBA BIL	Permanent working capital only; SBA memo and CAM process.	SBA-specific documentation and guarantee review.
Term Loan	May be secured or unsecured; fixed rate and repayment stipulations.	Term-loan specific underwriting and collateral check.
Group Credit	Tier I: BCC, TL, BIL, BALOC actual loans. Tier II examples: merchant guidance, cash-management ACH, purchasing.	Compute and evidence both group-credit amounts; clarify when Tier II overrides Tier I.

Unit warning: The screenshots literally show $100M and $250M, and the strategy chart shows $100M. Verify whether these are truly millions or represent another system unit before coding. The document does not silently convert them.

6. BALOC restrictions and exception checks

• SBA and conventional BALOC cannot be changed into one another on the same line; require a new line.

• Conventional secured/unsecured and express/standard may be changed, subject to collateral retention when reducing a secured line below $100M.

• A conventional line scoring SBA may require a larger requested line for SBA treatment; this is a policy-specific note requiring confirmation.

• Line increases are restricted within 6 months of opening per product-definition screenshot; decline-reasons screenshot says increases not allowed within 12 months. Conflict: obtain policy-owner resolution; do not hard-code either as approved.

• Investment real estate company BALOC cap shown as $25M. Startup BALOC cap shown as $25M revolving; term loans not subject to this stated startup BALOC cap.

7. Quick App knockout criteria (from underwriting strategy image)

Condition	Screenshot interpretation	System action for POC
Average personal FICO < 650	Decline	DECLINE_CANDIDATE; confirm borrower vs guarantor score selection.
SBSS < 160	Decline	DECLINE_CANDIDATE.
Negative EBITDA	Knockout criterion	DECLINE_CANDIDATE if approved EBITDA definition and period available.
Bankruptcy in last 5 years	Based on discharge date	DECLINE_CANDIDATE if verified date within policy window.
Applicant under 18	Knockout criterion	DATA_REVIEW / eligibility failure; confirm business vs individual applicant.
Maximum delinquency public records = 1, 2, 3	Listed as knockout; meaning of codes unspecified	POLICY_REVIEW until code definitions are provided.

Condition	Screenshot interpretation	System action for POC
SBSS 160-189; ownership < 2 years; FICO 650-759; request > $100M	Specific combined knockout shown	DECLINE_CANDIDATE only when all conditions match; confirm boundary/units.

8. Quick App decision tree (source-derived, nonbinding)

Prerequisites: identity joins valid, Quick App eligibility valid, required scores/ownership/NAICS available, and knockout checks completed.

SBSS branch	Additional tests	Strategy-chart outcome
<160	None	Decline
160-189	Ownership >=2 years	SBA approval path
160-189	Ownership <2 years, FICO >=760	SBA approval path
160-189	Ownership <2 years, FICO 650-759, requested <= $100M	SBA approval path
160-189	Ownership <2 years, FICO 650-759, requested > $100M	Decline
190-199	None	SBA approval path
>=200	Ownership <2 years	SBA approval path
>=200	Ownership >=2 years; high-risk NAICS	SBA guarantee path
>=200	Ownership >=2 years; other NAICS	Conventional approval path

Chart annotation: high-risk NAICS examples include 23 (Construction), 4451 (Grocery / Convenience Retailers), 45 (Retail trade / Amazon sellers), 8111 (Automotive Repair and Maintenance), 8121 (Personal Care Services), and 8123 (Drycleaning and Laundry Services). The chart states conventional approval overrides SBA nodes for vehicle-only loans. Do not treat these paths as authorized loan approvals.

9. Decline-reason taxonomy and data availability

Reason family	Evidence / logic from screenshots	Status
Bankruptcy / derogatory	Bankruptcy indicator and discharge date; public record patterns including judgements, liens, collections.	Implement with verified report/date.
Collections / judgement	Business: unsatisfied judgements or recent collections; screenshot notes manual review for some small older collections. Guarantor: unsatisfied judgements or collection indicator.	Needs exact age/amount thresholds.
Insufficient capital	Requested line relative to revenue; example text: 10% of revenue may drive line size.	Underwriter review; not a universal threshold.
Insufficient financial condition	YOY revenue decrease, expense increase, shrinking profits, negative EBITDA.	Implement when periods defined.
Limited credit experience	Accounts < 3 or thin-file indicator true.	Review / decline candidate by approved policy.
Delinquency / credit	Recent delinquencies, slow payments, credit concentrations, tax liens.	Requires bureau definitions and thresholds.

Reason family	Evidence / logic from screenshots	Status
Unable to verify identity / income	New-to-bank identity or income cannot be verified.	DATA_REVIEW / authorized underwriting.
Collateral / loan terms	Insufficient collateral, outside footprint, asset age vs loan term.	Manual review / out of POC where indicated.
Out of scope for POC	Negative news, lawsuits, garnishment, SBA eligibility, LTV and other screenshot-marked categories.	Report NOT_EVALUATED; never claim clear.

Other screenshot-listed decline reasons include excessive debt obligations, insufficient length of operations (<2 years), slow personal credit pay, foreclosure/repo, broker-referred applications, business model concerns, inadequate bank-line paydown, and loan-purpose concerns. Several are marked out of scope or not currently available in the system.

10. Rule ordering, precedence, and policy provenance

• Stage 0: schema, foreign keys, source freshness and borrower/guarantor mapping.

• Stage 1: policy approval status and POC mode; production cannot proceed without signed, versioned policy.

• Stage 2: product/Quick App eligibility and document completeness.

• Stage 3: hard knockouts (OFAC flag must trigger compliance review/hold; do not equate an unverified match with confirmed sanction), credit freeze, legal eligibility and other verified stop conditions.

• Stage 4: credit/financial scoring, ownership and NAICS decision tree.

• Stage 5: route to SIMULATED_APPROVAL_CANDIDATE, SIMULATED_DECLINE_CANDIDATE, MANUAL_REVIEW, FULL_UNDERWRITING or DATA_REVIEW.

• Stage 6: generate explanation with evidence references and missing-input list.

Policy conflicts: Original draft guide proposed OFAC -> DENY, credit freeze -> MANUAL_REVIEW, multiple bankruptcies -> DENY/REVIEW, high past due -> REVIEW and strong SBSS -> APPROVE candidate. The newer screenshots provide more detailed decision criteria. Resolve conflicts with policy/compliance before production; a possible OFAC match is a compliance hold pending confirmation, not an automatically substantiated denial.

11. Rule catalog: implementation examples

Rule ID	Input	Test / action
MTB-IDENTITY	account_id, loan_id, entity involvement, borrower role	Unresolved join or ambiguous borrower -> DATA_REVIEW.
MTB-ELIG-QUICK	requested_amount, tier1_group_credit, nonprofit	Quick App only if all source criteria satisfied.
MTB-KO-FICO	personal_fico	<650 -> simulated decline candidate.
MTB-KO-SBSS	sbss_score	<160 -> simulated decline candidate.
MTB-KO-EBITDA	ebitda	Negative -> simulated decline candidate.
MTB-KO-BANKRUPTCY	discharge_date, evaluation_date	Within 5 years -> simulated decline candidate.
MTB-STRATEGY	sbss, ownership_years, fico, request, naics	Apply Section 8 decision tree.

Rule ID	Input	Test / action
MTB-FREEZE	credit_report_freeze	Hold for review; cannot treat absent report as clean.
MTB-OFAC	ofac_alert	Potential match -> compliance review; verified status required.
MTB-POLICY-SIGNOFF	policy_version, owner approval, effective date	No operational authorization without approval.

12. Expected machine-readable decision contract

Recommended response example (synthetic):

```
{
  "loan_id": "LOAN-0001",
  "account_id": "ACC-0001",
  "mode": "SIMULATION",
  "eligibility_route": "QUICK_APP",
  "policy_route": "SIMULATED_APPROVAL_CANDIDATE",
  "authorized_decision": "NOT_ISSUED",
  "policy_version": "DRAFT-POC-1",
  "rule_results": [
    {"id":"MTB-KO-FICO","status":"PASS","value":735},
    {"id":"MTB-STRATEGY","status":"MATCH","branch":"SBSS>=200"}
  ],
  "missing_inputs": [],
  "source_records": ["Q1:LOAN-0001", "Q2:IND-0001", "Q4:CR-0001"],
  "human_review_required": true,
  "explanation": "Synthetic test case; not a credit offer."
}
```

13. LLM responsibilities, prompt controls and auditability

• Supply only the canonical structured record and deterministic rule outcomes to the model, not raw unvalidated spreadsheets as policy authority.

• Prompt: explain triggered and non-triggered rules with factual evidence, identify missing information, and never invent thresholds, scores, relationships or approval authority.

• Constrain JSON output with a schema; reject mismatched route or rule IDs and retry once, otherwise fall back to a deterministic explanation.

• Store policy version, input hash, report timestamps, rule evaluations, LLM model/version, prompt version, reviewer ID and decision history.

• Separate adverse-action reason codes from free-text narrative; legally required notices and fair-lending controls need compliance review.

14. Validation / acceptance tests

Test	Expected result
Strong SBSS, ownership >=2 years, low-risk NAICS, no blockers	SIMULATED_APPROVAL_CANDIDATE; no authorized approval.
FICO 640	SIMULATED_DECLINE_CANDIDATE with knockout evidence.
SBSS 155	SIMULATED_DECLINE_CANDIDATE.
Credit freeze Y	MANUAL_REVIEW / unable to evaluate score reliably.

Test	Expected result
Unresolved borrower vs guarantor join	DATA_REVIEW with exact missing foreign keys.
OFAC alert Y	COMPLIANCE_REVIEW, not automatic confirmed sanction.
Ownership 1 year, SBSS 175, FICO 700, amount over chart limit	Simulated decline branch, if units and rules confirmed.
Nonprofit Quick App	FULL_UNDERWRITING / not eligible for Quick App.
Policy draft / unsigned	Simulation allowed only; operational authorization blocked.
Conflicting BALOC increase windows	POLICY_REVIEW, not silently chosen threshold.

15. Decisions required before production

• Confirm all field API names and provide the original MTB Background Info workbook, not screenshots alone.

• Resolve money units ($100M, $250M), requested amount definition, and Tier I vs Tier II exposure calculation.

• Resolve BALOC line-increase 6-month vs 12-month conflict.

• Confirm personal FICO aggregation (average across which applicants/guarantors), SBSS response selection and report recency.

• Define the meaning of delinquency public-record codes 1/2/3, negative EBITDA timeframe, collections thresholds and bankruptcy date logic.

• Approve NAICS list, vehicle-only override, SBA guarantee criteria, decline-reason mappings and compliance handling.

• Obtain formal credit-policy, compliance, model-risk and legal sign-off; validate performance, fairness and explanation quality before any live decision.

Document status: Detailed implementation guideline reconstructed from supplied screenshots. It is not a substitute for the approved underwriting policy, system-of-record schema, or formal sign-off.

16. Complete decline-reason inventory transcribed from screenshots

The following inventory captures all visible rows from the Decline Reasons screenshots. "Not used", "out of scope" and "manual review" are retained as written; listing a reason does not authorize an automatic denial.

Decline reason	Source description / evidence	Implementation status
Adverse Media/Derogatory Public Information	Negative news hit / red flags; refer to full negative news memo	Out of POC; external evidence
Affiliated Businesses with Excessive Debt Obligation	Affiliate exposure and unsecured leverage; judgmental review	Conditional review
Age of corresponding asset does not match requested length of loan	Collateral age and useful life vs amortization term	Collateral review
Bankruptcy	Bankruptcy indicator and discharge date; last 5 years mentioned	Knockout candidate
Borrower Business Model Concerns - Insufficient Demonstration of Market Demand	Not used by Business Banking	Not used

Decline reason	Source description / evidence	Implementation status
Borrower Business Model Concerns - Market Saturation	Not used by Business Banking	Not used
Broker referred applications not accepted at M&T	Broker fee populated or broker source	Referral policy
Business expenses exceed business income	Negative EBITDA; use SBSS earnings before interest and taxes where EBITDA unavailable	Knockout candidate
Closed for Incompleteness	Incomplete application / clock expiry	Administrative
Collection action or judgment	Unsatisfied judgments, recent collections; borrower and guarantor logic differs	Knockout/review; thresholds missing
Deficient working capital	Short-term obligations exceed short-term liquidity; current assets less current liabilities	Financial review
Derogatory business credit	Multiple late payments, judgments, liens, collections and patterns	Credit review
Derogatory personal credit	Maximum delinquency 1,2,3 in last 12 months; 0 investigated	Credit review
Deteriorating Financial Market Conditions	Not used for Business Banking	Not used
Excessive Debt Obligations	Affiliates, borrower and guarantors; judgmental	Conditional review
Existing Credit Concentrations in Market	Out of scope for POC	Out of POC
Excessive debt obligations in relation to income	Negative EBITDA	Financial review
Existing M&T Bank line does not reflect adequate principal paydowns	Not used in Business Banking	Not used
Foreclosure or repossession	No system data point; external public records	Out of POC
Garnishment	Personal garnishment on credit report or other source	External evidence
Heightened Level of Risk	Conventional line increases; behavioral-to-behavioral PD grade	Specific use case
Incomplete information/credit application	Full underwriting missing information	Administrative review
Increase not allowed on accounts opened within the past 12 months	BALOC only; conflicts with separate 6-month note	Policy conflict
Insufficient capital for amount of credit requested	Gross annual revenues and line size; 10% revenue illustrative	Judgmental review
Insufficient financial condition	YOY revenues falling, expenses rising, profits shrinking	Financial review
Insufficient length of business operations	Business formation date less than 2 years	Conditional review
Intended use of funds is not appropriate for requested product/terms	Out of scope for POC	Out of POC
Interest rate included on Term sheet is too low	Out of scope for POC	Out of POC
Lack of established earnings record	Out of scope for POC	Out of POC

Decline reason	Source description / evidence	Implementation status
Limited credit experience	Number of accounts less than 3 OR thin-file indicator true	Credit review
Loan to Value Ratio	Out of scope for POC; larger underwriting	Out of POC
Outside of market area	Collateral / governing law state outside footprint	Geographic review
Pending Lawsuit	Not available on system; external source needed	Out of POC
SBA Ineligible	Outside scope of POC; product eligibility possible	Out of POC
Slow business credit pay	Business report payment trends	Credit review
Slow or past due in trade or loan payments	Borrower past due trade, balances and slow pay	Credit review
Slow personal credit pay	Guarantor late trade payments	Credit review
Tax liens or judgements	Filing dates, amounts, unsatisfied counts, tax lien indicator	Credit review
Transaction Does Not Meet Risk/Return Thresholds	Not used in Business Banking	Not used
Unable to Verify Identity/Business or your Intent to Apply	New-to-bank fraud and verification checks	Identity review
Unable to verify income	New-to-bank; cannot verify gross annual revenues	Income review
Value or type of collateral not sufficient	Underwriter / collateral value and type	Collateral review
No Reason: Credit Counseling Indicator	Out of scope for POC	Out of POC

17. Source field dictionary and API mapping

Names below are transcribed from the visible Field Mapping sheet; the original workbook is required to verify exact case and underscore counts. Rows without legible API names are identified as such.

Source entity	API name (verify)	Meaning
Loan	ID	Loan record ID
Loan	LLC_BI__Product_Line__c	Product line
Loan	LLC_BI__Product_Type__c	Product type
Loan	LLC_BI__Product__c	Product
Loan	LLC_BI__Amount__c	Loan/commit amount
Loan	SBSS7__annual_interest_expense__c	Annual interest expense
Loan	SBSS7__cash_and_equiv__c	Cash and equivalents
Loan	SBSS7__curr_dda_balance__c	Current DDA balance
Loan	SBSS7__earn_before_int_and_taxes__c	EBIT
Loan	SBSS7__net_worth__c	Net worth

Source entity	API name (verify)	Meaning
Loan	SBSS7__total_assets__c	Total assets
Loan	SBSS7__total_curr_assets__c	Current assets
Loan	SBSS7__total_curr_liabilities__c	Current liabilities
Loan	SBSS7__total_liabilities__c	Total liabilities
Loan	LLC_BI__Governing_Law_State__c	Governing law state
Loan Extension	Behavioral_Underwriting_Program__c	Behavioral underwriting program
Account	Id	Profile ID
Account	Name	Profile name
Account	Estimated_Annual_Revenue__c	Prior-year gross revenue
Account	Business_Net_Income__c	Business net income
Account	Birth_Date__c	Birth date
Account	Type	Profile type
Account	Years_Under_Ownership	Ownership years; API name verify
Account	Organization_Established_Year__c	Established year
Account	NAICS_Code_Formula__c	NAICS code
Account	Group_Credit_Amount__c	Group credit
Credit report	nFUSE__Account__c	Account FK
Credit report	nFUSE__Credit_Report_Freeze__c	Credit freeze
Credit report	nFUSE__Credit_Score__c	FICO/score
Credit report	Number_of_Public_Records__c	Public records
Credit report	Now_Delinquent__c	Current delinquency
Credit report	nFUSE__Total_Past_Due__c	Total past due
Credit report	nFUSE__Status__c	Report status
Credit report	nFUSE__Tax_Lien_Indicator__c	Tax lien
Credit report	nFUSE__Total_Credit_Limit__c	Credit limit
Credit report	nFUSE__Report_Category__c	Report category
Credit report	nFUSE__No_Hit__c	No hit
Credit report	nFUSE__Number_Of_Accounts__c	Number accounts
Credit report	nFUSE__Number_Of_Bankruptcies__c	Bankruptcies
Credit report	nFUSE__Number_Of_Charge_Offs__c	Charge offs

Source entity	API name (verify)	Meaning
Credit report	nFUSE__Number_Of_Judgements__c	Judgments
Credit report	nFUSE__Number_Of_Judgements_Satisfied__c	Satisfied judgments
Credit report	nFUSE__Number_Of_Liens__c	Liens
Credit report	nFUSE__Number_Of_Liens_Satisfied__c	Satisfied liens
Credit report	nFUSE__OFAC_Alert_Indicator__c	OFAC alert
Credit report	nFUSE__Open_Tradelines__c	Open trade lines
Credit report	nFUSE__Recent_Bankruptcy_Filing_Date__c	Bankruptcy filing date
Credit report	nFUSE__Recent_Judgements_Filing_Date__c	Judgment date
Credit report	nFUSE__Recent_Liens_Filing_Date__c	Lien date
Credit report	nFUSE__Charge_Off__c	Charge off
Credit report	nFUSE__Collection_Indicator__c	Collection
Credit report	nFUSE__Available_Credit__c	Available credit
Credit report	nFUSE__Average_Open_Balance__c	Average balance
Credit report	nFUSE__Bankruptcy_Amount__c	Bankruptcy amount
Credit report	nFUSE__Bankruptcy_Discharged__c	Bankruptcy discharged
Credit report	Critical_Alerts__c	Critical alerts
LCResponse	SCOC__SBSS7Score__c	SBSS score
Legal entity	LLC_BI__Borrower_Type__c	Borrower or guarantor
Indicator	SCOC__IndicatorKey__c	Key
Indicator	SCOC__IndicatorValue__c	Value
Characteristic	SCOC__CharacteristicKey__c	Key
Characteristic	SCOC__CharacteristicValue__c	Value

18. Characteristic and indicator key catalog

Visible characteristic keys from the field mapping screenshot. Definitions are not supplied; do not infer the semantics of abbreviated key names.

avgMosInFile, avgMosInFileTLRptd0To2Mos, highestUtilOnBankNatlRevTL, lowestRating, lowestRatingIL, lowestRatingOpndLast12Mos, lowestRatingOpndLast24Mos, lowestRatingOpndLast36Mos

lowestRatingRevOpen, maxDelqEver, maxDelqPR0To11Mos, mosSncMostRcnt30pDelq, mosSncMostRcnt60pDelq, mosSncMostRcntDtOpnd, mosSncMostRcntFinTLOpnd, mosSncMostRcntInq

mosSncOldestBankNatlRevOpenTLOpnd, mosSncOldestDtOpnd, netFrctIL, netFrctMtg, netFrctRev, numBankNatlRevTL90PctRptd0To2Mos, numBankNatlRevTLWBal75PctAmt, numCollection

numDaysInq0to11MosExclLast30Days, numFinTL, numInq0to11MosExclLast30Days, numInq0to5MosExclLast14Days, numInq0to5MosExclLast7Days, numPR, numRevOpenTLWBal, numRevTL30pDaysEver

numRevTLWBal50PctAmt, numTL, numTL30pDaysEverDerogPR, numTL60pDaysEverDerogPR, numTL90pDaysEverDerogPR, numTLOpnd3MosAndNotGT2x30Days, pctTLNeverDelq, pctTLNowCurr

totBal, totRevBal

Indicator example: thinFileIndicator=Y or N. Other indicator keys may exist in the full source export. The source screenshots do not provide a complete controlled vocabulary or interpretation for each characteristic.

19. Detailed Quick App decision table (source diagram)

Priority / branch	Condition	Simulated path
Precheck	FICO below 650	Decline candidate
1	SBSS below 160	Decline candidate
2a	SBSS 160-189 and ownership at least 2 years	SBA approval candidate
2b	SBSS 160-189, ownership below 2, FICO at least 760	SBA approval candidate
2c	SBSS 160-189, ownership below 2, FICO 650-759, requested amount at most $100M (units unverified)	SBA approval candidate
2d	SBSS 160-189, ownership below 2, FICO 650-759, requested amount above $100M (units unverified)	Decline candidate
3	SBSS 190-199	SBA approval candidate
4a	SBSS at least 200, ownership below 2 years	SBA approval candidate
4b	SBSS at least 200, ownership at least 2, high-risk NAICS	SBA approval candidate
4c	SBSS at least 200, ownership at least 2, non-high-risk NAICS	Conventional approval candidate

Diagram annotation: conventional approval overrides SBA nodes for vehicle-only loans. High-risk NAICS examples shown: 23 Construction, 4451 Grocery/Convenience Retailers, 45 Retail Trade/Amazon Retailers, 8111 Automotive Repair and Maintenance, 8121 Personal Care Services, 8123 Drycleaning and Laundry Services. Verify classification granularity and the SBA guarantee condition.

20. Full rule-to-source traceability

Rule ID	Source attributes	Outcome / validation
ID-01	Loan.Account FK; legal entity loan_id/account_id/role	Data review on invalid join; guarantor distinct from borrower
EL-01	Requested amount; Tier I/II group credit; Account.Type	Quick App eligibility / full underwriting
EL-02	Product line/type; security; collateral	Express vs conventional BALOC / term loan routing
EL-03	Account age; increase request	BALOC 6/12 month conflict => policy review
KO-01	Credit report OFAC flag	Compliance hold for possible match
KO-02	Credit report freeze	Manual review

Rule ID	Source attributes	Outcome / validation
KO-03	Credit score by eligible persons; aggregation method	FICO below 650 candidate; aggregation unresolved
KO-04	LCResponse SBSS7 score	Below 160 candidate
KO-05	SBSS EBIT; financial period	Negative earnings candidate; EBITDA mapping unresolved
KO-06	Bankruptcy discharge date and report date	Within five years candidate; precise date rules unresolved
KO-07	Max delinquency/public-record characteristic	1,2,3; exact semantics require validation
KO-08	Applicant DOB and reference date	Under 18; applicability to business/guarantor unresolved
STR-01	SBSS, ownership, FICO, amount, NAICS	Decision tree Section 19
CR-01	Judgments, satisfied counts, collections, liens	Borrower vs guarantor logic, age/amount thresholds
CR-02	Accounts count, thinFileIndicator	Limited credit experience
FIN-01	Current assets/current liabilities	Deficient working capital
FIN-02	Revenue, requested line, existing exposure	Insufficient capital judgmental
POL-01	Approved policy ID, version, effective dates	Operational decision blocked without sign-off

21. End-to-end synthetic acceptance scenarios

Case	Setup	Expected test outcome
T01	FICO 735; SBSS 215; ownership 5 years; low-risk NAICS; valid joins	Conventional approval candidate in simulation
T02	FICO 645; all other inputs healthy	Decline candidate FICO
T03	FICO 710; SBSS 150	Decline candidate SBSS
T04	FICO 720; SBSS 180; ownership 1 year; amount above diagram limit	Decline candidate branch
T05	FICO 780; SBSS 180; ownership 1 year	SBA approval candidate
T06	FICO 720; SBSS 195	SBA approval candidate
T07	FICO 735; SBSS 210; ownership 3; high-risk NAICS	SBA approval candidate
T08	FICO 735; SBSS 210; ownership 3; low-risk NAICS	Conventional approval candidate
T09	Credit freeze Y	Manual review
T10	OFAC alert Y	Compliance review; not automatically confirmed match
T11	Loan ID matches but legal-entity involvement absent	Data review with missing FK
T12	Guarantor FICO low but borrower FICO strong	Role-aware aggregation; policy review until rule defined

Case	Setup	Expected test outcome
T13	Bankruptcy discharge 4 years before evaluation	Decline candidate if approved policy confirms
T14	Nonprofit under Quick App limits	Full underwriting
T15	BALOC increase at 8 months	Policy conflict; 6 vs 12 months
T16	All good inputs; policy unsigned	Simulation only; authorized decision not issued
T17	Query 2 indicator Y and N for same entity/timestamp	Data conflict, no silent selection
T18	No credit-report timestamp	Data review; cannot claim clean report

22. Technical implementation checklist

• Create a relational staging model for Account, Loan, EntityInvolvement, CreditReport, LCResponse, LCREntity, DataSourceAnalysis, Indicator and Characteristic.

• Build a source dictionary with API names, source objects, field types, required flags, and timestamp precedence; reject ambiguous keys.

• Use per-loan canonical JSON with nested participating accounts, reports and evidence pointers; keep one-to-many records without row explosion.

• Implement rule registry with ID, scope, predicate, version, source, precedence, missing-data action, exception path and decline reason.

• Record per-rule PASS / FAIL / NOT_EVALUATED / REVIEW and separate eligibility_route, simulated_credit_route and authorized_decision.

• Generate model explanations only from verified feature values and deterministic rule results; prevent the LLM from changing decision codes.

• Create unit tests for every decision-tree branch, negative condition, null, contradictory source, and policy conflict.

• Obtain policy owner sign-off for thresholds, SBA eligibility, high-risk NAICS list, adverse-action reason codes, and lending governance.

Coverage boundary: This specification incorporates all legible screenshot rows and key structures supplied in the conversation. It is not a verified full export of the source workbook. Some field spellings, coded values, policy applicability and unseen rows may differ. Validate against the native MTB Background Info workbook and approved credit policy.